BDA Final — Mock Paper 1 Solutions

Worked solutions with mark allocation

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Topics covered in this PDF:

- row A: 1 cdot 0
- row B: 0
- row C: 0
- row D: 0

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Prepared as a comprehensive professor-grade study tutor.

Contents

1. Solution A1 [Bonferroni — 5 marks]
2. Solution A2 [Apriori — 5 marks]
3. Solution A3 [MinHash — 5 marks]
4. Solution A4 [Cosine Similarity — 5 marks]
5. Solution B1 [User-User CF — 10 marks]
6. Solution B2 [PageRank with Dead-End — 10 marks]
7. Solution B3 [k-Means — 10 marks]
8. Solution C1 [Topic-Sensitive PR + Spam Mass — 25 marks]
9. Solution C2 [Community Detection — 25 marks]

How to use this file.

First attempt the paper *unaided* under timed conditions. Only then read the solutions. For every part you missed, write the *exact* mistake on a separate sheet — most marks lost on this examiner come from arithmetic slips and missing formulas, not from conceptual gaps.

Mark allocation philosophy.

This examiner gives **method marks** liberally. Throughout these solutions, the per-line mark hints (e.g. *[1 mark — formula]*, *[1 mark — substitution]*) reflect how he typically allocates partial credit. Even on a wrong final answer, you keep all method marks if you wrote the correct intermediate steps.

Solution A1 [Bonferroni — 5 marks]

(a) Expected false positives. *[2 marks]*

$$\mathbb{E}[\text{FP}] = N \cdot p = 200{,}000 \times 10^{-4} = 20$$

So we expect **20 false-positive flags per day**.

(b) Real-fraud fraction. *[1 mark]*

True positives ≈ 8 per day. Total flagged ≈ false + true = $20 + 8 = 28$. Fraction real = $8 / 28 \approx 0.286$ — about **29% of all flagged transfers are genuine laundering**, the other 71% are noise.

(c) Bonferroni-corrected threshold. *[2 marks]*

$$\alpha_{\text{per-test}} = \frac{\alpha}{N} = \frac{0.05}{200{,}000} = 2.5 \times 10^{-7}$$

The original $p = 10^{-4}$ is **400× too loose** for the family-wise rate to be controlled. Either tighten the per-test threshold to $le\ 2.5 \times 10^{-7}$, or pre-filter N down to a much smaller candidate set.

Note

Marker note. Award 2 marks for $N \cdot p = 20$, 1 mark for any reasonable comparison answer in (b), and 2 marks for the correct Bonferroni number with a comment.

Solution A2 [Apriori — 5 marks]

(a) Singleton supports and L_1. *[2 marks]*

Item	Baskets containing it	σ
A	T_1, T_2, T_3, T_5	4
B	T_1, T_2, T_4, T_5	4

Item	Baskets containing it	σ
C	T_1, T_3, T_4, T_5	4
D	T_2, T_3, T_4, T_5	4
E	(none)	0

With threshold $s = 3$: $L_1 = \{A, B, C, D\}$. Item E is dropped.

(b) C_2, support, and L_2. *[3 marks]*

Join + prune from L_1 produces every pair of frequent singletons (no pruning shrinks C_2 here because every $(k-1) = 1$-subset is in L_1):

$$C_2 = \{\{A,B\}, \{A,C\}, \{A,D\}, \{B,C\}, \{B,D\}, \{C,D\}\}$$

Count support by scanning every basket:

Pair	Baskets containing it	σ	Frequent ?
$\{A,B\}$	T_1, T_2, T_5	3	✓
$\{A,C\}$	T_1, T_3, T_5	3	✓
$\{A,D\}$	T_2, T_3, T_5	3	✓
$\{B,C\}$	T_1, T_4, T_5	3	✓
$\{B,D\}$	T_2, T_4, T_5	3	✓
$\{C,D\}$	T_3, T_4, T_5	3	✓

$$L_2 = \{\{A,B\}, \{A,C\}, \{A,D\}, \{B,C\}, \{B,D\}, \{C,D\}\} \quad \text{(all six pairs frequent)}$$

Note

Marker note. 1 mark for L_1, 1 mark for naming the 6 candidates correctly, 2 marks for the support counts and final L_2. No mark deduction if the student also lists E in C_2 candidates as long as they correctly drop it for failing the join condition.

Solution A3 [MinHash — 5 marks]

Hash values for each row:

Row r	$h_1(r) = r \bmod 5$	$h_2(r) = (2r + 3) \bmod 5$
1	1	5 mod 5 = 0
2	2	7 mod 5 = 2
3	3	9 mod 5 = 4
4	4	11 mod 5 = 1

(a) Row-by-row trace. *[4 marks]*

Initialize M_{sig} to all ∞:

	C_1	C_2	C_3
h_1	∞	∞	∞
h_2	∞	∞	∞

Row *r=1* (h_1=1, h_2=0). $C_1 = 1$ → update C_1. $C_2 = 0$ → skip. $C_3 = 1$ → update C_3.

	C_1	C_2	C_3
h_1	1	∞	1
h_2	0	∞	0

Row *r=2* (h_1=2, h_2=2). $C_1 = 0$ → skip. $C_2 = 1$ → update. $C_3 = 1$ → $min(1, 2) = 1$, $min(0, 2) = 0$ (no change).

	C_1	C_2	C_3
h_1	1	2	1
h_2	0	2	0

Row *r=3* (h_1=3, h_2=4). $C_1 = 1$ → $min(1,3) = 1$, $min(0,4) = 0$ (no change). $C_2 = 1$ → $min(2,3) = 2$, $min(2,4) = 2$ (no change). $C_3 = 0$ → skip.

	C_1	C_2	C_3
h_1	1	2	1
h_2	0	2	0

Row *r=4* (h_1=4, h_2=1). $C_1 = 0$ → skip. $C_2 = 1$ → $min(2,4) = 2$, $min(2,1) = 1$ (update h_2). $C_3 = 1$ → $min(1,4) = 1$, $min(0,1) = 0$ (no change).

Final signature matrix:

	C_1	C_2	C_3
h_1	1	2	1
h_2	0	1	0

(b) Estimate $J(C_1, C_3)$. *[1 mark]*

The two signatures of C_1 and C_3 are identical: *(1, 0)* and *(1, 0)*. So both rows agree:

$$\hat{J}(C_1, C_3) = \frac{\#\text{matching rows}}{\#\text{rows}} = \frac{2}{2} = 1.0$$

(With only $n = 2$ hash functions this estimator has very high variance; the true Jaccard from the membership matrix is $J(C_1, C_3) = |\{1\}|/|\{1, 2, 3, 4\}| = 1/4 = 0.25$, but the question asks for the *estimate from the signatures*, which is 1.0.)

> **Note**
>
> *Marker note.* 4 marks for any 4 of (initialise to ∞, correct row-1 trace, correct row-2 trace, correct row-3 trace, correct row-4 trace, correct final matrix). 1 mark for the Jaccard estimate. Forgetting to initialise to ∞ is the most common error and costs 1–2 marks.

Solution A4 [Cosine Similarity — 5 marks]

(a) Raw cosine. *[4 marks]*

Vectors: $u = (4, 5, 2, 3)$, $v = (5, 3, 4, 4)$.

Step 1 — dot product. *[1 mark]*

$$u \cdot v = (4)(5) + (5)(3) + (2)(4) + (3)(4) = 20 + 15 + 8 + 12 = 55$$

Step 2 — norms. *[2 marks]*

$$\|u\| = \sqrt{4^2 + 5^2 + 2^2 + 3^2} = \sqrt{16 + 25 + 4 + 9} = \sqrt{54} \approx 7.348$$

$$\|v\| = \sqrt{5^2 + 3^2 + 4^2 + 4^2} = \sqrt{25 + 9 + 16 + 16} = \sqrt{66} \approx 8.124$$

Step 3 — cosine. *[1 mark]*

$$\cos(u, v) = \frac{55}{7.348 \times 8.124} = \frac{55}{59.70} \approx 0.921$$

The two users are highly similar (raw cosine close to 1 because their rating vectors point in nearly the same direction in $mathbbR^4$).

(b) From cosine to Pearson. *[1 mark]*

Mean-centre each vector before computing cosine. Replace u with $u_c = u - \bar{u}\ 1$ and likewise $v_c = v - \bar{v}\ 1$, where $\bar{u} = (4+5+2+3)/4 = 3.5$ and $\bar{v} = (5+3+4+4)/4 = 4$. Then $Pearson(u, v) = cos(u_c, v_c)$. Pearson differs from cosine *only* by this centring step.

> **Note**
>
> *Marker note.* Full 4 marks for parts of (a) require: 1 for dot, 2 for both norms (1 each), 1 for the final ratio. The 1 mark in (b) is awarded for the keyword **mean-centring**.

Solution B1 [User-User CF — 10 marks]

(a) Co-rated items and Sara's mean. *[2 marks]*

Sara has rated $\{M_1, M_2, M_4, M_5\}$ (she did not rate M_3). Each of Ali, Bilal, Hina has rated all five movies, so Sara's set of co-rated items with each of them is $\{M_1, M_2, M_4, M_5\}$.

$$\bar{r}_{\text{Sara}} = (4 + 3 + 5 + 2)/4 = 14/4 = 3.5$$

So Sara's centred vector over (M_1, M_2, M_4, M_5) is

$$\text{Sara}_c = (4 - 3.5,\ 3 - 3.5,\ 5 - 3.5,\ 2 - 3.5) = (0.5,\ -0.5,\ 1.5,\ -1.5)$$

with $\|Sara_c\|^2 = 0.25 + 0.25 + 2.25 + 2.25 = 5.0$, so $\|Sara_c\| = \sqrt{5} \approx 2.236$.

(b) Pearson similarities. *[5 marks — about 1.5 marks each]*

Sara vs Ali over (M_1, M_2, M_4, M_5). Ali's ratings on those four $= (5, 3, 4, 2)$, mean $= 14/4 = 3.5$.

$$\text{Ali}_c = (1.5,\ -0.5,\ 0.5,\ -1.5)$$

Numerator:

$$5)(1.5) + (-0.5)(-0.5) + (1.5)(0.5) + (-1.5)(-1.5) = 0.75 + 0.25 + 0.75 + 2.25 = 4$$

$\|Ali_c\|^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5.0$, so $\|Ali_c\| = \sqrt{5} \approx 2.236$.

$$\text{sim}(\text{Sara}, \text{Ali}) = \frac{4.0}{\sqrt{5} \cdot \sqrt{5}} = \frac{4.0}{5.0} = +0.800$$

Sara vs Bilal. Bilal on $(M_1, M_2, M_4, M_5) = (3, 1, 3, 5)$, mean $= 12/4 = 3.0$. $Bilal_c = (0, -2, 0, +2)$.

Numerator: $(0.5)(0) + (-0.5)(-2) + (1.5)(0) + (-1.5)(2) = 0 + 1 + 0 - 3 = -2.0$. $\|Bilal_c\|^2 = 0 + 4 + 0 + 4 = 8$, so $\|Bilal_c\| = 2\sqrt{2} \approx 2.828$.

$$\text{sim}(\text{Sara}, \text{Bilal}) = \frac{-2.0}{\sqrt{5} \cdot 2\sqrt{2}} = \frac{-2.0}{\sqrt{40}} = \frac{-2.0}{6.325} \approx -0.316$$

Sara vs Hina. Hina on $(M_1, M_2, M_4, M_5) = (1, 5, 2, 1)$, mean $= 9/4 = 2.25$. $Hina_c = (-1.25, 2.75, -0.25, -1.25)$.

Numerator: $(0.5)(-1.25) + (-0.5)(2.75) + (1.5)(-0.25) + (-1.5)(-1.25) = -0.625 - 1.375 - 0.375 + 1.875 = -0.500$.

$\|Hina_c\|^2 = 1.5625 + 7.5625 + 0.0625 + 1.5625 = 10.75$, so $\|Hina_c\| = \sqrt{10.75} \approx 3.279$.

$$\text{sim}(\text{Sara}, \text{Hina}) = \frac{-0.500}{\sqrt{5} \cdot \sqrt{10.75}} = \frac{-0.500}{7.331} \approx -0.068$$

Summary:

Neighbour	Pearson sim	$	\text{sim}	$
Ali	*+0.800*	0.800		
Bilal	*-0.316*	0.316		
Hina	*-0.068*	0.068		

(c) Top-2 neighbours. *[2 marks]*

By absolute similarity: **Ali** (*0.800*) and **Bilal** (*0.316*).

(d) Predict Sara's rating for M_3. *[1 mark]*

Use the weighted-average formula with the two neighbours' actual M_3 ratings $r(Ali, M_3) = 4$ and $r(Bilal, M_3) = 2$:

$$\hat{r}(\text{Sara}, M_3) = \frac{(0.800)(4) + (-0.316)(2)}{0.800 + 0.316} = \frac{3.200 - 0.632}{1.116} = \frac{2.568}{1.116} \approx 2.30$$

Predicted rating ≈ 2.3 stars.

> **Note**
>
> *Marker note.* For (b), award 1.5 marks per similarity (1 for the centred vectors and numerator, 0.5 for the final ratio). Award full marks even if signs are accidentally swapped, provided the final top-2 in (c) is consistent. The negative sim for Bilal *correctly* enters the predictor with a minus sign.

Solution B2 [PageRank with Dead-End — 10 marks]

(a) Build *M*. *[3 marks]*

Out-degrees: $d_A = 2$ (out-links to B, C), $d_B = 1$ (to C), $d_C = 1$ (to A), $d_D = 0$ (**dead-end**). Column D is therefore **all zeros**.

	from *A*	from *B*	from *C*	from *D*
row *A*	0	0	1	0
row *B*	1/2	0	0	0
row *C*	1/2	1	0	0
row *D*	0	0	0	0

Column sums: A: $1/2 + 1/2 = 1$ ✓. B: 1 ✓. C: 1 ✓. D: 0 □ — that's the dead-end signature.

(b) Three iterations. *[6 marks — 2 each]*

Iteration 1. $r^{(0)} = (0.25, 0.25, 0.25, 0.25)$^→$p$.

$M\, r^{(0)}$ row-by-row:

- row A: $1 \cdot 0.25 = 0.250$
- row B: $0.5 \cdot 0.25 = 0.125$
- row C: $0.5 \cdot 0.25 + 1 \cdot 0.25 = 0.375$
- row D: 0

$\beta\, M\, r^{(0)} = 0.8 \cdot (0.250, 0.125, 0.375, 0) = (0.200, 0.100, 0.300, 0.000)$. $S = 0.200 + 0.100 + 0.300 + 0 = 0.600$. Leak $1 - S = 0.400$. Per-node leak $= 0.400/4 = 0.100$. $r^{(1)} = (0.200 + 0.100, 0.100 + 0.100, 0.300 + 0.100, 0 + 0.100) = (0.300, 0.200, 0.400, 0.100)$. **Sum = 1.000** ✓.

Iteration 2. $r^{(1)} = (0.300, 0.200, 0.400, 0.100)$.

$M\, r^{(1)}$:

- row A: $1 \cdot 0.400 = 0.400$
- row B: $0.5 \cdot 0.300 = 0.150$
- row C: $0.5 \cdot 0.300 + 1 \cdot 0.200 = 0.350$
- row D: 0

$\beta\, M\, r^{(1)} = (0.320, 0.120, 0.280, 0.000)$. $S = 0.720$. Leak $= 0.280$. Per-node $= 0.070$. $r^{(2)} = (0.390, 0.190, 0.350, 0.070)$. **Sum = 1.000** ✓.

Iteration 3. $r^{(2)} = (0.390, 0.190, 0.350, 0.070)$.

$M\, r^{(2)}$:

- row A: $1 \cdot 0.350 = 0.350$
- row B: $0.5 \cdot 0.390 = 0.195$
- row C: $0.5 \cdot 0.390 + 1 \cdot 0.190 = 0.385$
- row D: 0

$\beta M r^{(2)} = (0.280, 0.156, 0.308, 0.000)$. $S = 0.744$. Leak = 0.256. Per-node = 0.064. $r^{(3)} = (0.344, 0.220, 0.372, 0.064)$. **Sum = 1.000** ✓.

(c) Highest / lowest rank, dead-end behaviour. *[1 mark]*

After iteration 3, **highest = *C* (0.372)**, **lowest = *D* (0.064)**. Node *D* has no in-links (no node points to *D* in this graph), so the *only* rank flowing into *D* is the uniform leak-redistribution term *(1-S)/N*. With every iteration *D* receives exactly *(1-S)/N* but contributes nothing back to the system — its rank is effectively pinned to that small share.

> **Note**
>
> *Marker note.* Award 2 marks per iteration for correct r'^{new}, leak, and final r^{new} with sum-check. Forgetting to redistribute the leak is the single most common error — the symptom is sums dropping below 1.

Solution B3 [k-Means — 10 marks]

(a) Iteration 1 — assignments. *[4 marks]*

Initial centroids $c_1 = (2, 2)$, $c_2 = (7, 7)$.

Point	$d^2(\cdot, c_1)$	$d^2(\cdot, c_2)$	Nearer	Assigned
$P_1 = (1,1)$	$(1)^2 + (1)^2 = 2$	$(6)^2 + (6)^2 = 72$	c_1	Cluster 1
$P_2 = (2,1)$	$0 + 1 = 1$	$25 + 36 = 61$	c_1	Cluster 1
$P_3 = (1,2)$	$1 + 0 = 1$	$36 + 25 = 61$	c_1	Cluster 1
$P_4 = (8,8)$	$36 + 36 = 72$	$1 + 1 = 2$	c_2	Cluster 2
$P_5 = (9,8)$	$49 + 36 = 85$	$4 + 1 = 5$	c_2	Cluster 2
$P_6 = (8,9)$	$36 + 49 = 85$	$1 + 4 = 5$	c_2	Cluster 2

So Cluster 1 = \$P_1$, P_2, P_3\, Cluster 2 = \$P_4$, P_5, P_6\.

(b) Recompute centroids. *[3 marks]*

$$c_1^{(1)} = \frac{(1+2+1,\ 1+1+2)}{3} = \left(\frac{4}{3},\ \frac{4}{3}\right) \approx (1.333,\ 1.333)$$

$$c_2^{(1)} = \frac{(8+9+8,\ 8+8+9)}{3} = \left(\frac{25}{3},\ \frac{25}{3}\right) \approx (8.333,\ 8.333)$$

(c) Iteration 2 — re-assign. *[2 marks]*

Every point in Cluster 1 lies near *(1, 1)*, much closer to $c_1^{(1)} \approx (1.33, 1.33)$ than to $c_2^{(1)} \approx (8.33, 8.33)$. By symmetry every Cluster-2 point is closer to $c_2^{(1)}$. Spot-check the closest

case:

$$d^2(P_2, c_1^{(1)}) = (2 - 1.333)^2 + (1 - 1.333)^2 \approx 0.444 + 0.111 = 0.556$$

$$d^2(P_2, c_2^{(1)}) = (2 - 8.333)^2 + (1 - 8.333)^2 \approx 40.11 + 53.78 = 93.89$$

So P_2 stays in Cluster 1. By the same logic every other point keeps its cluster. **No point changes membership → centroids will not update further → the algorithm has converged.**

(d) Final clusters. *[1 mark]*

- Cluster 1 = \P_1, P_2, P_3\, centroid *(4/3, 4/3) ≈ (1.333, 1.333)*.
- Cluster 2 = \P_4, P_5, P_6\, centroid *(25/3, 25/3) ≈ (8.333, 8.333)*.

> **Note**
>
> *Marker note.* In (a), squared distances suffice — students who use √ at every step waste time and risk arithmetic slips. Full credit either way. In (c), the convergence justification (no membership change ⇒ centroids stable ⇒ done) is the key concept.

Solution C1 [Topic-Sensitive PR + Spam Mass — 25 marks]

(a) Build *M*. *[4 marks]*

Out-degrees: $d_1 = 2$ (to 2, 3), $d_2 = 1$ (to 3), $d_3 = 1$ (to 1), $d_4 = 2$ (to 1, 5), $d_5 = 1$ (to 4).

	from 1	from 2	from 3	from 4	from 5
row 1	0	0	1	1/2	0
row 2	1/2	0	0	0	0
row 3	1/2	1	0	0	0
row 4	0	0	0	0	1
row 5	0	0	0	1/2	0

Column sums: *1, 1, 1, 1, 1* ✓. No dead-ends.

(b) Two iterations of Topic-PR. *[8 marks — 4 per iteration]*

$v_S = (1, 0, 0, 0, 0)$^→p since $|S| = |\1\| = 1$. $\beta = 0.85$, $1 - \beta = 0.15$.

Iteration 1. $r^{(0)} = (0.20, 0.20, 0.20, 0.20, 0.20)$^→p.

$M\, r^{(0)}$ row-by-row:

- row 1: $1 \cdot r_3 + 0.5 \cdot r_4 = 0.20 + 0.10 = 0.300$
- row 2: $0.5 \cdot r_1 = 0.100$
- row 3: $0.5 \cdot r_1 + 1 \cdot r_2 = 0.100 + 0.200 = 0.300$
- row 4: $1 \cdot r_5 = 0.200$
- row 5: $0.5 \cdot r_4 = 0.100$

Sum = *0.300 + 0.100 + 0.300 + 0.200 + 0.100 = 1.000* ✓ (column-stochastic, no leak).

$\beta\, M\, r^{(0)} = 0.85 \cdot (0.300, 0.100, 0.300, 0.200, 0.100) = (0.255, 0.085, 0.255, 0.170, 0.085)$.

Add $(1-\beta)\, v_S = (0.150, 0, 0, 0, 0)$:

$$r^{(1)} = (0.405,\ 0.085,\ 0.255,\ 0.170,\ 0.085)$$

Sum = $0.405 + 0.085 + 0.255 + 0.170 + 0.085 = 1.000$ ✓.

Iteration 2. $r^{(1)} = (0.405, 0.085, 0.255, 0.170, 0.085)$.

$M\, r^{(1)}$:

- row 1: $r_3 + 0.5\, r_4 = 0.255 + 0.085 = 0.340$
- row 2: $0.5 \cdot r_1 = 0.5 \cdot 0.405 = 0.2025$
- row 3: $0.5 \cdot r_1 + r_2 = 0.2025 + 0.085 = 0.2875$
- row 4: $r_5 = 0.085$
- row 5: $0.5 \cdot r_4 = 0.5 \cdot 0.170 = 0.085$

Sum = $0.340 + 0.2025 + 0.2875 + 0.085 + 0.085 = 1.000$ ✓.

$\beta\, M\, r^{(1)} = 0.85 \cdot (0.340, 0.2025, 0.2875, 0.085, 0.085) = (0.289, 0.172, 0.244, 0.072, 0.072)$.

Add $(0.150, 0, 0, 0, 0)$:

$$r^{(2)} = (0.439,\ 0.172,\ 0.244,\ 0.072,\ 0.072)$$

Sum ≈ 1.000 ✓ (small rounding error in the last decimal).

(c) Compare to vanilla PageRank. *[4 marks]*

For vanilla PR, $v = (0.2, 0.2, 0.2, 0.2, 0.2)$ (uniform teleport). Reusing $M\, r^{(0)}$ from above:

$$\begin{aligned} r^{(1)}_{\text{vanilla}} &= 0.85 \cdot (0.300, 0.100, 0.300, 0.200, 0.100) + 0.15 \cdot (0.2, 0.2, 0.2, 0.2, 0.2) \\ &= (0.255, 0.085, 0.255, 0.170, 0.085) + (0.030, 0.030, 0.030, 0.030, 0.030) \\ &= (0.285,\ 0.115,\ 0.285,\ 0.200,\ 0.115) \end{aligned}$$

Comparison after 1 iteration:

Node	Topic-PR	Vanilla PR	Δ (topic − vanilla)
1	0.405	0.285	**+0.120**
2	0.085	0.115	−0.030
3	0.255	0.285	−0.030
4	0.170	0.200	−0.030
5	0.085	0.115	−0.030

Node 1 gains all of the teleport mass (the entire $(1-\beta) = 0.15$ instead of $0.15/5 = 0.03$), while every other node loses its 0.03 share. After several more iterations the teleport-into-1 mass also propagates to nodes 1's neighbours (here: 2 and 3) more than to the disconnected component $\{4, 5\}$. The final stationary distribution will be heavily concentrated on $\{1, 2, 3\}$.

(d) Spam Mass. *[5 marks]*

Apply $SM(p) = (r_p - r^+_p)/r_p$ to each node:

Node	r	r^+	$r - r^+$	$SM = (r-r^+)/r$
1	0.30	0.40	*-0.10*	*-0.10/0.30 ≈ -0.333*
2	0.20	0.15	*+0.05*	*0.05/0.20 = 0.250*
3	0.25	0.25	*0*	*0/0.25 = 0.000*
4	0.15	0.10	*+0.05*	*0.05/0.15 ≈ 0.333*
5	0.10	0.05	*+0.05*	*0.05/0.10 = 0.500*

Spam threshold *SM ge 0.4* flags only node 5 (*SM = 0.50*). Nodes 1, 2, 3, 4 are below threshold and considered legitimate.

Note on node 1's negative SM: node 1 receives *more* TrustRank than vanilla PR — that means trusted seeds disproportionately reach node 1, making it strongly trusted. The negative number is a feature, not a bug.

(e) Effect of lowering β. *[4 marks — 1 each]*

Going $\beta = 0.85 \rightarrow 0.50$ shifts the random surfer's behaviour from "follow links 85% of the time" to "follow links 50% / teleport 50%." Effects:

1. **Rank of node 1 *increases* further.** With $|S| = 1$ at node 1, lowering β raises the teleport coefficient $(1-\beta)$ from 0.15 to 0.50 — node 1 now gets $+0.50$ injected per iteration directly. Topic-PR concentrates *more* mass at the seed when β shrinks.
2. **Rank of nodes far from node 1 (e.g. 4, 5) *decreases*.** They receive contributions only via link-following from node 1's neighbourhood, which is dampened by β. Lower β means less rank propagates outward, so distant nodes lose mass.
3. **Convergence speeds up.** The iteration is a contraction with rate β; smaller $\beta \Rightarrow$ faster convergence (the spectral gap is $1 - \beta$). At $\beta = 0.5$, errors halve every step.
4. **Spam-farm robustness *increases*.** The spam-farm multiplier $1/(1 - \beta^2)$ shrinks: at $\beta = 0.85$ it is ≈ 3.6, at $\beta = 0.5$ it is ≈ 1.33. Spammers can no longer boost their target as much.

> **Note**
>
> *Marker note.* In (b), give 4 marks per iteration: 1 for the Mr row-by-row, 1 for the $\beta M r$ multiplication, 1 for the teleport addition, 1 for the sum-check. In (e), award 1 mark for each of the four sub-bullets; correct direction (up/down/faster) is enough — exact magnitudes not required.

Solution C2 [Community Detection — 25 marks]

(a) Brandes BFS from source 1. *[8 marks]*

BFS levels and shortest-path counts σ:

Level	Node	σ	Predecessor(s) in BFS DAG
0	1	1	(root)

Level	Node	σ	Predecessor(s) in BFS DAG
1	2	1	1
1	3	1	1
2	4	1	3
3	5	1	4
3	6	1	4

DAG edges (direction = "downward in level"): *1→2, 1→3, 3→4, 4→5, 4→6*. Edges *2-3* and *5-6* are intra-level → **not in the DAG**, so they receive **0** credit from this source.

Brandes credit propagation (bottom-up). Each node starts with self-credit 1; each DAG edge from child *c* to parent *p* receives $credit(c) \cdot \sigma(p)/\sigma(c)$, then *p* accumulates that flow.

Step	Node *c* (level)	*credit(c)*	DAG edge to parent	Edge contribution
1	5 (L3)	1	4–5	$1 \cdot \sigma(4)/\sigma(5) = 1$
2	6 (L3)	1	4–6	$1 \cdot \sigma(4)/\sigma(6) = 1$
3	4 (L2)	$1 + 1 + 1 = 3$	3–4	$3 \cdot \sigma(3)/\sigma(4) = 3$
4	2 (L1)	$1 + 0 = 1$	1–2	$1 \cdot \sigma(1)/\sigma(2) = 1$
5	3 (L1)	$1 + 3 = 4$	1–3	$4 \cdot \sigma(1)/\sigma(3) = 4$

Edge-betweenness contributions from source *s* = *1*:

Edge	Contribution from *s* = *1*
1–2	1
1–3	4
2–3	0 (not in DAG)
3–4	3
4–5	1
4–6	1
5–6	0 (not in DAG)

(The total betweenness summed across all sources, divided by 2, gives the global table provided in part (b): the bridge edge 3–4 wins with $b = 9$.)

(b) Remove highest-betweenness edge and recompute. *[5 marks]*

From the supplied table, the highest is **edge 3–4 with *b* = *9***. Removing it disconnects the graph into two components:

- Component 1: triangle on *1, 2, 3*\ with edges *1-2, 1-3, 2-3*\.
- Component 2: triangle on *4, 5, 6*\ with edges *4-5, 4-6, 5-6*\.

Recompute betweenness within each component. In a triangle, every pair of nodes is joined by a direct edge of length 1; there are also length-2 alternate paths through the third node, but they are not shortest. So each of the three pairs contributes 1 to its corresponding direct edge:

Edge (in remaining graph)	*b(e)* on the post-cut graph
1–2	1
1–3	1
2–3	1
4–5	1
4–6	1
5–6	1

All six remaining edges are tied at $b = 1$. (The single bridge has done all the heavy lifting; once it is gone, no edge stands out.)

(c) Modularity. *[8 marks]*

Compute on the **original** graph (before any cut). $m = 7$ edges, so $2m = 14$.

Step 1 — degrees. *[2 marks]*

Node	Neighbours	k_i
1	2, 3	2
2	1, 3	2
3	1, 2, 4	3
4	3, 5, 6	3
5	4, 6	2
6	4, 5	2

Check: $\Sigma_i k_i = 2 + 2 + 3 + 3 + 2 + 2 = 14 = 2m$ ✓.

Step 2 — community statistics for C_1 = \1,2,3\ and C_2 = \4,5,6\. *[3 marks]*

Internal edges of C_1: edges with both endpoints in \1, 2, 3\ = \1-2, 1-3, 2-3\ → $L_1 = 3$. Sum of degrees: $D_1 = k_1 + k_2 + k_3 = 2 + 2 + 3 = 7$.

Internal edges of C_2: edges with both endpoints in \4, 5, 6\ = \4-5, 4-6, 5-6\ → $L_2 = 3$. $D_2 = k_4 + k_5 + k_6 = 3 + 2 + 2 = 7$.

(The bridge edge 3–4 has one endpoint in C_1 and one in C_2 — it is *not* internal to either community, so it does **not** count toward L_1 or L_2.)

Step 3 — apply the block formula. *[3 marks]*

$$Q = \sum_{c \in \{C_1, C_2\}} \left[\frac{L_c}{m} - \left(\frac{D_c}{2m} \right)^2 \right]$$

$$Q = \left[\frac{3}{7} - \left(\frac{7}{14}\right)^2\right] + \left[\frac{3}{7} - \left(\frac{7}{14}\right)^2\right] = 2 \cdot \left[\frac{3}{7} - \frac{1}{4}\right]$$
$$= 2 \cdot \left[\frac{12}{28} - \frac{7}{28}\right] = 2 \cdot \frac{5}{28} = \frac{10}{28} = \frac{5}{14} \approx 0.357$$

Q ≈ 0.36 — strong community structure (typical "good" partitions score *0.3–0.7*).

(d) Laplacian and the Fiedler vector. *[4 marks]*

Degree matrix *D = diag(2, 2, 3, 3, 2, 2)*. Adjacency:

A = beginpmatrix 0 & 1 & 1 & 0 & 0 & 0 1 & 0 & 1 & 0 & 0 & 0 1 & 1 & 0 & 1 & 0 & 0 0 & 0 & 1 & 0 & 1 & 1 0 & 0 & 0 & 1 & 0 & 1 0 & 0 & 0 & 1 & 1 & 0 endpmatrix

Laplacian *L = D - A*:

L = beginpmatrix 2 & -1 & -1 & 0 & 0 & 0 -1 & 2 & -1 & 0 & 0 & 0 -1 & -1 & 3 & -1 & 0 & 0 0 & 0 & -1 & 3 & -1 & -1 0 & 0 & 0 & -1 & 2 & -1 0 & 0 & 0 & -1 & -1 & 2 endpmatrix

Verify rows sum to 0: row 3: *-1 - 1 + 3 - 1 = 0* ✓; row 4: *-1 + 3 - 1 - 1 = 0* ✓; etc.

Why 1_6 is always an eigenvector. Each row of *L* sums to zero by construction ($L_{ii} = k_i$ on the diagonal; $L_{ij} = -1$ for each of the k_i neighbours of *i*). Hence $L \cdot 1_6 = 0$, i.e. 1_6 is an eigenvector with **eigenvalue $\lambda_1 = 0$**. (This is the trivial "everything in one cluster" mode.)

Fiedler vector and how it would partition this graph. The **Fiedler vector** v_2 is the eigenvector associated with the second-smallest eigenvalue $\lambda_2 > 0$. Sign-thresholding the entries of v_2 gives a 2-way cut: positive entries form one community, negative entries the other. For this barbell graph, v_2 would have:

- **Positive entries on \1, 2, 3** (one triangle),
- **Negative entries on \4, 5, 6** (the other triangle),
- entries closer to zero at the bridge endpoints (nodes 3 and 4) — they sit on the boundary.

This recovers exactly the same partition that Girvan-Newman gives by removing edge 3–4.

> **Note**
>
> *Marker note.* In (a), award 4 marks for correct BFS levels and *σ* values, and 4 marks for the credit propagation table. In (c), heavily reward the $\Sigma k_i = 2m$ check — students who skip it often plug D_c into the wrong place. In (d), do *not* deduct marks for not actually computing λ_2; the question only requires writing *L* and explaining the Fiedler partition qualitatively.

After grading yourself against these solutions, fill in the table below to direct your revision:

Question	Marks earned	Marks possible	Topic	Action if < 80%
A1		5	Bonferroni	Re-read W01 §3 + WE 1
A2		5	Apriori	Re-read W02-03 §5 + WE 2
A3		5	MinHash	Re-read W04-05 §6 trace

Question	Marks earned	Marks possible	Topic	Action if < 80%
A4		5	Cosine	Re-read W07 §5
B1		10	User-User CF	Re-read W07 WE 2
B2		10	PageRank dead-end	Re-read W08-09 §9 + WE 7
B3		10	k-Means	Re-read W13-14 §3
C1		25	Topic-PR + Spam Mass	Re-read W12 §3, §6, §7
C2		25	GN + Spectral	Re-read W16 §3, §4, §5, §7
Total		**100**		

Aim for 65/100.

If you scored below 50, focus the next study session on the lowest-scoring section. If you scored 65+ on the first attempt, move on to Mock Paper 2 (different datasets, same difficulty) and then to past-paper drills.

End of Mock Paper 1 — Worked Solutions.